Fourier-Transform:- (Time Domain → freq. Domain) Date ....

$x(n) \longrightarrow X(\omega)$ ← Continous Signal

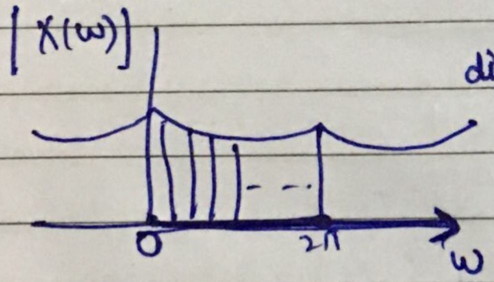
Discrete Sequence or Signal

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

But for ~~......~~ various applications of DSP such as Linear filtering, speech Processing, Image Processing there is requirement of frequency domain signal of discrete nature.

So, we need to convert $X(\omega)$ into a discrete sequence. Sampling is one of the solutions

for Example



magnitude Spectrum of DTFT

if we take N samples at diff. values of k. then freq. 2π will be divided by N.

So $\omega = \frac{2\pi}{N} \cdot K$ where $K = 0, 1, \ldots N-1$

So DFT of sequence $x(n)$ is given by:-

$$\boxed{X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N} k \cdot n}}$$

Date ....

Q.1 Compute the N-point DFT of the signal given by $x(n) = \delta(n)$.

Sol^n:

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn}$$

where $k = 0, \ldots N-1$
$n = 0, \ldots N-1$

$$X(k) = \sum_{n=0}^{N-1} \delta(n)\, e^{-j\frac{2\pi}{N}kn}$$

As $\boxed{\delta(n) = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}}$

$$= \delta(0)\, e^{-j\frac{2\pi}{N}k \cdot 0}$$

$$= \delta(0)\, e^{0}$$

$$\boxed{X(k) = 1}$$

Q.2 Compute the N-point DFT of the signal given by

$x(n) = \delta(n - n_0)$ for $0 < n_0 < N$

Sol^n

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} \delta(n - n_0)\, e^{-j\frac{2\pi}{N}kn}$$

$\boxed{\text{Putting } n = n_0}$

$$X(k) = \delta(0)\, e^{-j\frac{2\pi}{N}kn_0}$$

$$\boxed{X(k) = e^{-j\frac{2\pi}{N}kn_0}}$$

Date ....

Q3 Compute N-Point DFT of given signal,

$x(n) = a^n \quad ; \quad 0 \leq n \leq N-1$

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn}$$

$$X(k) = \sum_{n=0}^{N-1} a^n\, e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} \left(a\, e^{-j\frac{2\pi}{N}k}\right)^n$$

As we know

$$\boxed{\sum_{N_1}^{N_2} a^n = \frac{a^{N_1} - a^{N_2+1}}{1-a} \quad \text{for } a \neq 1}$$

$$X(k) = \frac{1 - \left[a\, e^{-j\frac{2\pi}{N}k}\right]^N}{1 - \left(a\, e^{-j\frac{2\pi}{N}k}\right)}$$

$$= \frac{1 - a^N e^{-j2\pi k}}{1 - \left(a\, e^{-j\frac{2\pi}{N}k}\right)}$$

$$e^{-j2\pi k} = \cos(2\pi k) - j\sin(2\pi k)$$

As $k = 0, 1, 2 \ldots$

$$\boxed{e^{-j2\pi k} = 1}$$

$$\boxed{X(k) = \frac{1 - a^N}{1 - a e^{-j\frac{2\pi}{N}k}}}$$

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Q.4 find the DFT of the sequence $x(n) = 1; 0 \le n \le 2$ for $N=4$. Sketch the magnitude & phase spectrum

Sol^n

for $N=4$ $\quad x(n) = \{\overset{0}{1}, \overset{1}{1}, \overset{2}{1}, \overset{3}{0}\}$

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}kn}$$

where $k = 0, 1 \ldots N-1$

$$X(k) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{2\pi}{4}kn} \quad \leftarrow \boxed{\text{As } N=4}$$

& $k = 0, 1, 2, 3$

$$X(k) = x(0)e^{0} + x(1)e^{-j\frac{2\pi}{4}k\cdot 1} + x(2)e^{-j\frac{2\pi}{4}k\cdot 2} + x(3)e^{-j\frac{2\pi}{4}\cdot k\cdot 3}$$

~~X(k) = 1~~

for $k=0$

$$X(0) = 1 + 1\cdot e^{-j\frac{2\pi}{4}\cdot 0\cdot 1} + 1\,e^{-j\frac{2\pi}{4}\cdot 0\cdot 2} + 0\cdot e^{-j\frac{2\pi}{4}\cdot 0\cdot 3}$$

$$X(0) = 1 + 1 + 1 + 0$$

$$\boxed{X(0) = 3}$$

for $k=1$

$$X(1) = 1 + 1\cdot e^{-j\frac{2\pi}{4}\cdot 1} + 1\cdot e^{-j\frac{2\pi}{4}\cdot 1\cdot 2} + 0$$

$$= 1 + e^{-j\frac{\pi}{2}} + 1\,e^{-j\pi}$$

$$e^{-j\frac{\pi}{2}} = \cos\theta - j\sin\theta$$

$$= \cos\frac{\pi}{2} - \sin\frac{\pi}{2} = 0 - j$$

$$\boxed{e^{-j\frac{\pi}{2}} = -j}$$

$$e^{-j\pi} = \cos\pi - j\sin\pi$$

$$e^{-j\pi} = -1$$

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$X(1) = \cancel{1} - j - \cancel{1}$

$\boxed{X(1) = -j}$

for k=2 $\quad X(2) = 1 + 1 \cdot e^{-j\frac{2\pi}{4} \cdot 2} + 1 \cdot e^{-j\frac{2\pi}{4} \cdot 2 \cdot 2} + 0$

$X(2) = 1 + e^{-\pi} + e^{-j2\pi}$

$X(2) = 1 - 1 + 1$

$\boxed{X(2) = 1}$

for k=3 $\quad X(3) = 1 + [1 \times j] + [1 \times (-1)] + 0$

$X(3) = \cancel{1} + j - \cancel{1}$

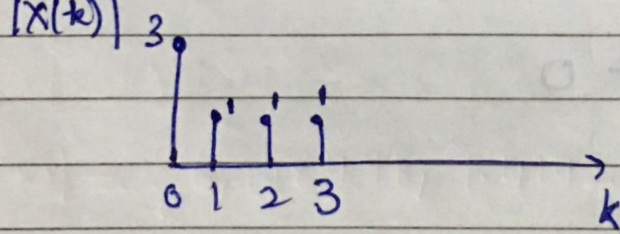
$\boxed{X(3) = j}$

$\therefore X(k) = \{3, -j, 1, j\}$

magnitude Spectrum:- $|X(k)| = \{3, 1, 1, 1\}$

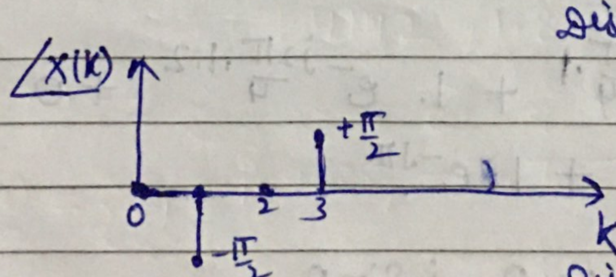
Phase Spectrum $\angle X(k) = \{0, -\frac{\pi}{2}, 0, \frac{\pi}{2}\}$

ms →



Discrete frequency

PS →



Discrete frequency

Date ....

Q.5 Compute the 8-Point DFT of the signal

$s(n) = \{1, 1, 1, 1, 1, 1, 0, 0\}$

Also sketch its magnitude & Phase Spectrum.

Sol^n DFT of signal $s(n)$ is given by:-

$$S(k) = \sum_{n=0}^{N-1} s(n)\, e^{-j\frac{2\pi}{N}kn}$$

As $\boxed{N=8}$

$$= \sum_{n=0}^{7} s(n)\, e^{-j\frac{2\pi}{8}kn}$$

$$S(k) = s(0)\, e^{0} + s(1)\, e^{-j\frac{2\pi}{8}k} + s(2)\, e^{-j\frac{2\pi}{8}k\cdot 2} + s(3)\, e^{-j\frac{2\pi}{8}k\cdot 3}$$
$$+ s(4)\, e^{-j\frac{2\pi}{8}k\cdot 4} + s(5)\, e^{-j\frac{2\pi}{8}k\cdot 5} + s(6)\, e^{-j\frac{2\pi}{8}k\cdot 6} + s(7)\, e^{-j\frac{2\pi}{8}k\cdot 7}$$

$$S(k) = 1 + 1\cdot e^{-j\frac{\pi}{4}k} + 1\cdot e^{-j\frac{\pi}{2}k} + 1\cdot e^{-j\frac{3\pi}{4}k} + 1\cdot e^{-j\pi k} + 1\cdot e^{-j\frac{5\pi}{4}k}$$
$$+ 0 + 0$$

$$S(k) = 1 + e^{-j\frac{\pi}{4}k} + e^{-j\frac{\pi}{2}k} + e^{-j\frac{3\pi}{4}k} + e^{-j\pi k} + e^{-j\frac{5\pi}{4}k}$$

for k=0

$$S(0) = 1 + e^{0} + e^{0} + e^{0} + e^{0} + e^{0} = 6$$

$\boxed{S(0) = 6}$

for k=1

$$S(1) = 1 + e^{-j\frac{\pi}{4}} + e^{-j\frac{\pi}{2}} + e^{-j\frac{3\pi}{4}} + e^{-j\pi} + e^{-j\frac{5\pi}{4}}$$

$\boxed{S(1) = -0.7071 - j\,1.7071}$

for k=2

$\boxed{S(2) = 1 - j}$

for k=3

$\boxed{S(3) = 0.7071 + j0.2929}$

for k=4

$\boxed{S(4) = 0}$

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for k=5

$$\boxed{S(5) = 0.7071 - j0.2929}$$

for k=6

$$\boxed{S(6) = 1+j}$$

for k=7

$$\boxed{S(7) = -0.7071 + j1.7071}$$

So DFT of S(n) is given by:—

$$S(K) = \{6, -0.7071 - j1.7071, 1-j, -0.7071 + j0.2929, 0, 0.7071 - j0.2029, 1+j, 0.7071 + j1.7071\}$$

$$\text{Amplitude} = \sqrt{(\text{Real Part})^2 + (\text{Imaginary Part})^2}$$

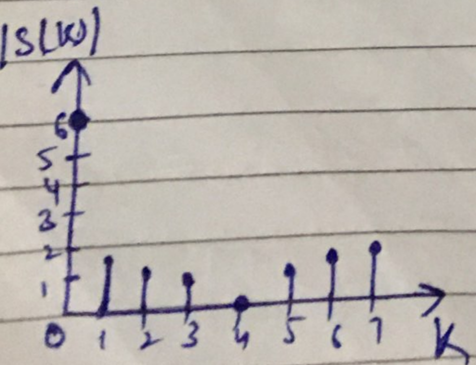
$$\text{Phase} = \tan^{-1}\left(\frac{\text{Imaginary Part}}{\text{Real Part}}\right)$$

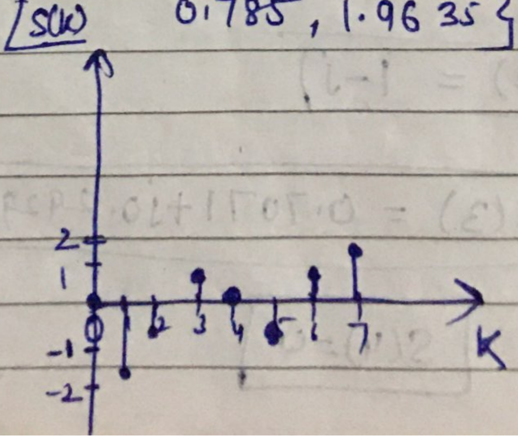
Amplitude of S(k):

$$|S(k)| = \{6, 1.8478, 1.4142, 0.7654, 0, 0.7654, 1.4142, 1.8478\}$$

Phase of S(K):—

$$\angle S(k) = \{0, -1.9635, -0.785, 0.3927, 0, -0.3927, 0.785, 1.9635\}$$





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Q.6 Compute 4-Point DFT of sequence given below:-

$$x(n) = \begin{cases} \frac{1}{3} & \text{for } 0 \le n \le 2 \\ 0 & \text{elsewhere} \end{cases} \qquad \text{for } N=4$$

So DFT of given sequence is:-

$k = 0, 1, 2, 3$

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N} k.n}$$

$$X(k) = \sum_{n=0}^{3} x(n)\, e^{-j\frac{\pi}{2} k.n}$$

$$X(K) = x(0)\, e^{0} + x(1)\, e^{-j\frac{\pi}{2}K} + x(2)\, e^{-j\pi k} + x(3) e^{-j\frac{3\pi}{2}}$$

$$= \frac{1}{3} + \frac{1}{3} \cdot e^{-j\frac{\pi}{2}k} + \frac{1}{3} e^{-j\pi k} + 0$$

$$\boxed{X(k) = \frac{1}{3} + \frac{1}{3} e^{j\frac{\pi}{2}k} + \frac{1}{3} e^{-j\pi k}}$$

for $k=0$

$$X(0) = \frac{1}{3} + \frac{1}{3} e^{0} + \frac{1}{3} e^{0} = 1$$

$$\boxed{X(0) = 1}$$

for $k=1$ $\quad X(1) = \frac{1}{3} + \frac{1}{3} e^{-j\frac{\pi}{2}} + \frac{1}{3} e^{-j\pi}$

$$X(1) = \cancel{\frac{1}{3}} + \frac{1}{3}(-j) + \cancel{\frac{1}{3}(-1)}$$

$$\boxed{X(1) = \frac{-j}{3}}$$

for $k=2$ $\quad \boxed{X(2) = \frac{1}{3}}$

for $K=3$ $\quad \boxed{X(3) = \frac{j}{3}}$

So $\left[X(k) = \left\{ 1, \frac{-j}{3}, \frac{1}{3}, \frac{j}{3} \right\} \right]$ **Ans**

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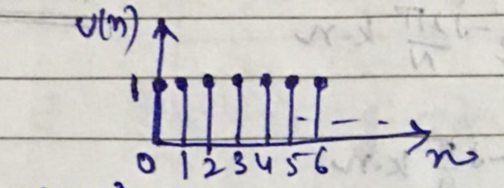
Q.7 Compute the DFT of following window function

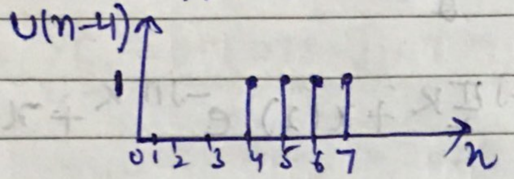
$$w(n) = u(n) - u(n-N)$$

Sol^n

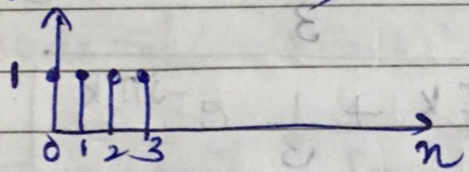
Let $N=4$

$$w(n) = u(n) - u(n-4)$$





$w(n) = u(n) - u(n-4)$



So $w(n) = \{\overset{0}{1}, \overset{1}{1}, \overset{2}{1}, \overset{3}{1}\}$ $\qquad \left\{\begin{array}{l} n = 0 \text{ to } N-1 \\ K = 0 \text{ to } N-1 \end{array}\right\}$

DFT

$$W(K) = \sum_{n=0}^{N-1} w(n)\, e^{-j\frac{2\pi}{N}Kn}$$

Put $N=4$

$$w(K) = \sum_{n=0}^{3} w(n)\, e^{-j\frac{2\pi}{4}K.n}$$

$$w(K) = \sum_{n=0}^{3} w(n)\, e^{-j\frac{\pi}{2}K.n}$$

$$w(K) = w(0)e^{0} + w(1)\, e^{-j\frac{\pi}{2}K} + w(2)\, e^{-j\frac{\pi}{2}.K.2} + w(3)e^{-j\frac{\pi}{2}K.3}$$

$$w(K) = 1 + 1 \cdot e^{-j\frac{\pi}{2}K} + 1 \cdot e^{-j\pi K} + 1 \cdot e^{-j\frac{3\pi}{2}K}$$

Scanned with CamScanner

As. $e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = -j$ & $e^{-j\frac{3\pi}{2}} = \cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2} = 0 - j(-1) = +j$

$e^{-j\pi} = \cos\pi - j\sin\pi = -1$ & $e^{-j2\pi} = \cos 2\pi - \sin 2\pi = 1$

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$$\boxed{W(k) = 1 + e^{-j\frac{\pi}{2}k} + e^{-j\pi k} + e^{-j\frac{3\pi}{2}k}}$$

for k=0

$$W(0) = 1 + e^{0} + e^{0} + e^{0}$$
$$= 1 + 1 + 1 + 1$$
$$\boxed{W(0) = 4}$$

for k=1

$$W(1) = 1 + e^{-j\frac{\pi}{2}} + e^{-j\pi} + e^{-j\frac{3\pi}{2}}$$
$$W(1) = 1 + (-j) + (-1) + j$$
$$W(1) = 1 - j - 1 + j = 0$$
$$\boxed{W(1) = 0}$$

for k=2

$$W(2) = 1 + e^{-j\pi.k} + e^{-2j\pi} + e^{-j3\pi}$$
$$W(2) = 1 + (-1) + 1 + (-1)$$
$$\boxed{W(2) = 0}$$

As $e^{-j3\pi} = \cos 3\pi - j\sin 3\pi = -1 - 0 = -1$

for k=3

$$W(3) = 1 + e^{-j\frac{3\pi}{2}} + e^{-j3\pi} + e^{-j\frac{9\pi}{2}}$$

$$W(3) = 1 + \left(\cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}\right) + \left(\cos 3\pi - j\sin 3\pi\right) + \left(\cos\frac{9\pi}{2} - j\sin\frac{9\pi}{2}\right)$$

$$W(3) = 1 + (0 + j) + (-1 - 0) + (0 - j)$$

$$W(3) = \cancel{1} + \cancel{j} - \cancel{1} - \cancel{j}$$
$$\boxed{W(3) = 0}$$

So $W(k) = \{4, 0, 0, 0\}$ Ans

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Q.7 find 6-Point DFT of $x(n) = \{1, 1, 0, 0, 2, 2\}$ using Linear Transformation or matrix method.

Sol^n

$$X_N = W_N x_N \quad \therefore N = 6$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \\ X(4) \\ X(5) \end{bmatrix} = \begin{array}{c|c} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} W_6^0 & W_6^0 & W_6^0 & W_6^0 & W_6^0 & W_6^0 \\ W_6^0 & W_6^1 & W_6^2 & W_6^3 & W_6^4 & W_6^5 \\ W_6^0 & W_6^2 & W_6^4 & W_6^6 & W_6^8 & W_6^{10} \\ W_6^0 & W_6^3 & W_6^6 & W_6^9 & W_6^{12} & W_6^{15} \\ W_6^0 & W_6^4 & W_6^8 & W_6^{12} & W_6^{16} & W_6^{20} \\ W_6^0 & W_6^5 & W_6^{10} & W_6^{15} & W_6^{20} & W_6^{25} \end{bmatrix} \end{array} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \end{bmatrix}$$

As we know Twiddle factor $W_N^K = e^{-j\frac{2\pi}{N}K}$

$W_6^0 = e^{-j\frac{2\pi}{N}\cdot 0} = e^0 = 1$

$W_6^1 = e^{-j\frac{2\pi}{6}} = \cos\left(\frac{2\pi}{6}\right) - j\sin\left(\frac{2\pi}{6}\right) = 0.5 - 0.866j$

$W_6^2 = -0.5 - 0.866j$

$W_6^3 = -1$

$W_6^4 = -0.5 + 0.886j$

$W_6^5 = 0.5 + 0.886j$

$W_6^6 = 1$

$W_6^8 = -0.5 - 0.866j$

$W_6^9 = -1$

$W_6^{10} = -0.5 + 0.886j$

$W_6^{12} = 1$

$W_6^{15} = -1$

$W_6^{16} = -0.5 + 0.886j$

$W_6^{20} = -0.5 - 0.886j$

$W_6^{25} = 0.5 - 0.886j$

Putting the values of all twiddle factors in the above matrix :—

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$$= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0.5-0.866j & -0.5-0.866j & -1 & -0.5+0.866j & 0.5+0.866j \\ 1 & -0.5-0.866j & -0.5+0.866j & 1 & -0.5-0.866j & -0.5+0.866j \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -0.5+0.886j & -0.5-0.866j & 1 & -0.5+0.886j & -0.5-0.886j \\ 1 & 0.5+0.886j & -0.5+0.886j & -1 & -0.5-0.886j & 0.5-0.886j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 0.5+0.866j \\ -0.5+0.866j \\ -2 \\ -0.5-0.866j \\ 0.5-0.866j \end{bmatrix}$$

So

$$X(K) = \{4,\ 0.5+0.866j,\ -0.5+0.866j,\ -2,\ -0.5-0.866j,\ 0.5-0.866j\}$$

Ans

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Q.8 Find 4-Point DFT of $x(n) = \{1, 0, 0, 1\}$ using Linear Transformation matrix method.

Sol^n

$$X_N = W_N x_N \qquad \therefore N = 4$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ 1 & W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ 2 & W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ 3 & W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{array} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

Values of Twiddle factor:-

$$W_N^k = e^{-j\frac{2\pi}{N}k} = e^{-j\frac{2\pi}{4}k}$$

$$W_4^0 = 1$$

$$W_4^1 = e^{-j\frac{2\pi}{4}\cdot 1} = \cos\left(\frac{2\pi}{4}\right) - j\sin\left(\frac{2\pi}{4}\right) = -j$$

$$W_4^2 = e^{-j\frac{2\pi}{4}\cdot 2} = \cos\pi - j\sin\pi = -1$$

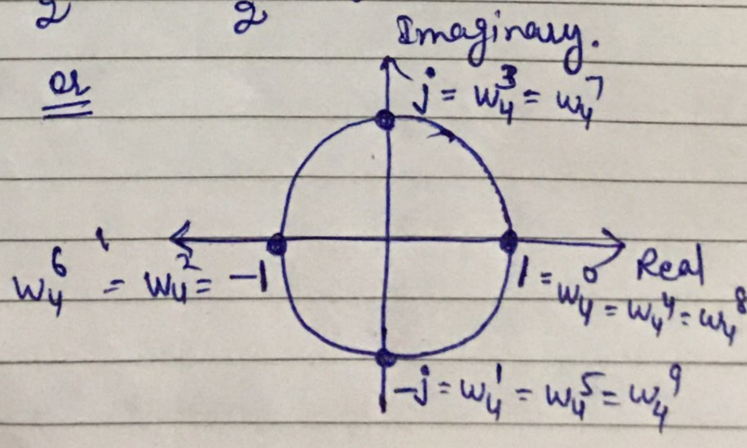
$$W_4^3 = e^{-j\frac{2\pi}{4}\cdot 3} = \cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2} = +j$$

$$W_4^4 = 1$$

$$W_4^6 = -1$$

$$W_4^9 = -j$$

or



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Putting the values of Twiddle factors:→

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & +j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1+j \\ 0 \\ 1-j \end{bmatrix}$$

So $\boxed{X(K) = \{2,\ 1+j,\ 0,\ 1-j\}}$ Ans.

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Q.9 find 8-Point DFT of $x(n) = \{1, 1, 2, 2, 3, 3, 4, 4\}$ Using matrix method.

$\boxed{N=8}$

Sol^n

$$X_N = W_N \cdot x_N$$

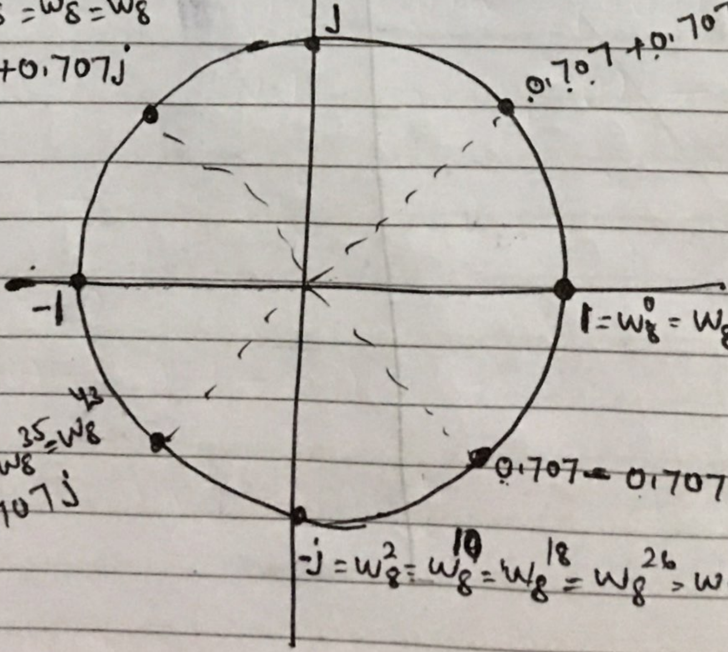
$$
\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \\ X(4) \\ X(5) \\ X(6) \\ X(7) \end{bmatrix}
=
\begin{array}{c|cccccccc}
 & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
\hline
0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 & W_8^0 \\
1 & W_8^0 & W_8^1 & W_8^2 & W_8^3 & W_8^4 & W_8^5 & W_8^6 & W_8^7 \\
2 & W_8^0 & W_8^2 & W_8^4 & W_8^6 & W_8^8 & W_8^{10} & W_8^{12} & W_8^{14} \\
3 & W_8^0 & W_8^3 & W_8^6 & W_8^9 & W_8^{12} & W_8^{15} & W_8^{18} & W_8^{21} \\
4 & W_8^0 & W_8^4 & W_8^8 & W_8^{12} & W_8^{16} & W_8^{20} & W_8^{24} & W_8^{28} \\
5 & W_8^0 & W_8^5 & W_8^{10} & W_8^{15} & W_8^{20} & W_8^{25} & W_8^{30} & W_8^{35} \\
6 & W_8^0 & W_8^6 & W_8^{12} & W_8^{18} & W_8^{24} & W_8^{30} & W_8^{36} & W_8^{42} \\
7 & W_8^0 & W_8^7 & W_8^{14} & W_8^{21} & W_8^{28} & W_8^{35} & W_8^{42} & W_8^{49}
\end{array}
\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \\ x(5) \\ x(6) \\ x(7) \end{bmatrix}
$$

$W_8^6 = W_8^{14} = W_8^{22} = W_8^{30} = W_8^{38} = W_8^{46}$

$W_8^5 = W_8^{13} = W_8^{21} = W_8^{29} = W_8^{37} = W_8^{45}$

$-0.707 + 0.707j$

$0.707 + 0.707j = W_8^7 = W_8^{15} = W_8^{23} = W_8^{31} = W_8^{39} = W_8^{47}$



$= W_8^{36} = W_8^{44}$

$W_8^{28} = W_8^{20} = W_8^{12} = W_8^4 = -1$

$1 = W_8^0 = W_8^8 = W_8^{16} = W_8^{24} = W_8^{32} = W_8^{40} = W_8^{48}$

$W_8^3 = W_8^{11} = W_8^{19} = W_8^{27} = W_8^{35} = W_8^{43}$

$= -0.707 - 0.707j$

$0.707 - 0.707j = W_8^1 = W_8^9 = W_8^{17} = W_8^{25} = W_8^{33}$

$-j = W_8^2 = W_8^{10} = W_8^{18} = W_8^{26} = W_8^{34} = W_8^{42}$ $= W_8^{41} = W_8^{49}$

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$$
\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \\ X(4) \\ X(5) \\ X(6) \\ X(7) \end{bmatrix}
=
\begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
1 & 0.707-0.707j & -j & -0.707-0.707j & -1 & -0.707+0.707j & j & 0.707+0.707j \\
1 & -j & -1 & j & 1 & -j & -1 & j \\
1 & -0.707+0.707j & j & 0.707-0.707j & -1 & 0.707+0.707j & -j & -0.707-0.707j \\
1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\
1 & -0.707-0.707j & -j & 0.707+0.707j & -1 & 0.707-0.707j & j & -0.707+0.707j \\
1 & j & -1 & -j & 1 & j & -1 & j \\
1 & 0.707+0.707j & j & -0.707-0.707j & -1 & -0.707+0.707j & -j & 0.707-0.707j
\end{bmatrix}
\begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \\ 3 \\ 3 \\ 4 \\ 4 \end{bmatrix}
$$

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$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \\ X(4) \\ X(5) \\ X(6) \\ X(7) \end{bmatrix} = \begin{bmatrix} 20 \\ -2+4.82j \\ -2+2j \\ -2+0.828j \\ 0 \\ -2-0.828j \\ -2-2j \\ -2-4.82j \end{bmatrix}$$

So $X(k) = \{20, -2+4.82j, -2+2j, -2+0.828j, 0, -2-0.828j, -2-2j, -2-4.82j\}$

Ans

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Q-10 for $X(k) = \{2, 1+j, 0, 1-j\}$, find 4-Point IDFT using matrix method.

Solⁿ

$$\boxed{x_N = \frac{1}{N}[W_N^*]X_N} \leftarrow \text{IDFT}$$

where W_N^* is complex conjugate of W_N.

$$\boxed{N=4}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & W_4^{*0} & W_4^{*0} & W_4^{*0} & W_4^{*0} \\ 1 & W_4^{*0} & W_4^{*1} & W_4^{*2} & W_4^{*3} \\ 2 & W_4^{*0} & W_4^{*2} & W_4^{*4} & W_4^{*6} \\ 3 & W_4^{*0} & W_4^{*3} & W_4^{*6} & W_4^{*9} \end{array} \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

As we know

$$W_N^K = e^{-j\frac{2\pi}{N}K}$$

So

$$W_N^{*K} = e^{+j\frac{2\pi}{N}K}$$

$W_4^{*0} = e^0 = 1$

$W_4^{*1} = e^{j\frac{2\pi}{4}} = e^{j\frac{\pi}{2}} = \cos\frac{\pi}{2} + j\sin\frac{\pi}{2} = j$

$W_4^{*2} = e^{j\frac{2\pi}{4}\cdot 2} = e^{j\pi} = \cos\pi + j\sin\pi = -1$

$W_4^{*3} = e^{j\frac{2\pi}{4}\cdot 3} = e^{j\frac{3\pi}{2}} = \cos\frac{3\pi}{2} + j\sin\frac{3\pi}{2} = 0 - j = -j$

$W_4^{*4} = e^{j\frac{2\pi}{4}\cdot 4} = e^{j2\pi} = \cos 2\pi + j\sin 2\pi = 1$

$W_4^{*6} = -1$

$W_4^{*9} = j$

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$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 2 \\ 1+j \\ 0 \\ 1-j \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 \\ 0 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

So $x(n) = \{1, 0, 0, 1\}$ Ans.

Q.11 Compute the length-4 sequence from its DFT using matrix method.

$$X(k) = \{4, 1-j, -2, 1+j\}$$

Solⁿ IDFT $= x(n) = \frac{1}{N}\left[W_N^{*}\right] X_N$

$$\boxed{N=4}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4} \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} W_4^{*0} & W_4^{*0} & W_4^{*0} & W_4^{*0} \\ W_4^{*0} & W_4^{*1} & W_4^{*2} & W_4^{*3} \\ W_4^{*0} & W_4^{*2} & W_4^{*4} & W_4^{*6} \\ W_4^{*0} & W_4^{*3} & W_4^{*6} & W_4^{*9} \end{bmatrix} \end{array} \begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 4 \\ 1-j \\ -2 \\ 1+j \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4+1-j-2+1+j \\ 4+j-j^2+2-j-j^2 \\ 4-1+j-2-1-j \\ 4-j+j^2+2+j+j^2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 \\ 4+2+1+1 \\ 4-4 \\ 4+2-2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 \\ 8 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

So $\boxed{x(n) = \{1, 2, 0, 1\}}$ Ans

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Q.12 Determine the Discrete fourier transform (DFT) of four point sequence. $x(n) = \{0, 1, 2, 3\}$

Solⁿ

$$X_N = [W_N]\, x(n)$$

$$X_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -j \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0+1+2+3 \\ 0-j-2+3j \\ 0-1+2-3j \\ 0+j-2-3j \end{bmatrix} = \begin{bmatrix} 6 \\ 2j-2 \\ -2 \\ -2j-2 \end{bmatrix}$$

So. $X(k) = \{\underset{\uparrow}{6}, 2j-2, -2, -2j-2\}$ Ans

Q.13 Determine DFT of a sequence $x(n) = \{1, 1, 0, 0\}$ and check the validity of your answer by calculating it's IDFT.

Solⁿ

$$X_N = [W_N]\, x(n)$$

$$\begin{matrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{matrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -j \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1-j \\ 0 \\ 1+j \end{bmatrix}$$

So $X(k) = \{2, 1-j, 0, 1+j\}$

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Now, let us check this answer by using the expression for IDFT.

$$\text{IDFT} \rightarrow x(n) = \frac{1}{N}\left[W_N^*\right] \cdot X_N$$

$$x(n) = \frac{1}{4}\begin{bmatrix} W_4^{*0} & W_4^{*0} & W_4^{*0} & W_4^{*0} \\ W_4^{*0} & W_4^{*1} & W_4^{*2} & W_4^{*3} \\ W_4^{*0} & W_4^{*2} & W_4^{*4} & W_4^{*6} \\ W_4^{*0} & W_4^{*3} & W_4^{*6} & W_4^{*9} \end{bmatrix}\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}\begin{bmatrix} 2 \\ 1-j \\ 0 \\ 1+j \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 2+1-j+0+1+j \\ 2+j+1+0-j+1 \\ 2-1+j+0-1-j \\ 2-j-1+0+j-1 \end{bmatrix}$$

$$x(n) = \frac{1}{4}\begin{bmatrix} 4 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{matrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{matrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

So $x(n) = \{1, 1, 0, 0\}$

this is same as given sequence. Hence, calculated DFT is correct.